

REVIEW

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Cocoa production in the 2020s: challenges and solutions

John Edem Kongor^{1*} , Margaret Owusu¹ and Charlotte Oduro-Yeboah¹

Abstract

Background Cocoa (*Theobroma cacao* L.) is a crop of huge economic significance worldwide and is grown mainly in tropical and subtropical countries. Currently, West Africa produces most of the world's cocoa. The crop provides economic support to cocoa-growing countries, smallholder farmers, and the chocolate confectionery industries. Cocoa is also valued for its appealing flavours and the health-promoting properties of the bioactive phytochemicals in the beans, which have received increased global attention in recent years.

Main body The cocoa industry is divided into two sectors: upstream (cocoa bean production and marketing), which is dominated by cocoa-producing countries, and downstream (cocoa bean processing activities to produce semi-finished and finished products). Pests and diseases, climate change, low soil fertility, high soil cadmium levels, and the ongoing Russian-Ukrainian conflict threaten the crop's long-term production. In addition to these challenges, cocoa cultivation also contributes to environmental and biodiversity degradation.

Conclusion To address these challenges and ensure a sustainable supply of high-quality cocoa beans to meet the rising global demand, sustainable intensification of its production in producing countries is deemed critical. These include breeding varieties that are resistant to yield-limiting factors, the use of integrated management strategies to improve soil fertility and control pests, diseases, and heavy metals like Cd, the implementation of agroforestry systems, increased farm gate prices, and the provision of social interventions such as alternative livelihoods for farmers to increase cocoa production on existing farmlands. Standardized and harmonized farm management and postharvest strategies are also required for the consistent production of high-quality beans each season.

Keywords Cocoa, Pests and diseases, Environment, Climate change, Sustainable intensification

Introduction

Cocoa (*Theobroma cacao* L.) is a tropical crop that belongs to the Malvaceae family and the genus *Theobroma* (Prabhakaran Nair 2010) and is commonly known as cacao or cocoa (Baharum et al. 2016). The *Theobroma* genus contains twenty-two (22) species, with *Theobroma cacao* L. being the most important and widely cultivated due to the value of its seeds (Kongor et al. 2016). Cocoa is

reported to be a native species of tropical humid forests on the lower eastern equatorial slopes of South America's Andes (Caligiani et al. 2016). Cacao is derived from the Olmec and subsequent Mayan languages (Kakaw), and cacahuatl is Nahuatl (Aztec language) derived from Olmec/Mayan etymology (Dillinger et al. 2000). Cocoa cultivation dates back to 600 AD in the lowlands of south Yucatan by the Maya (Beckett 2008). According to Purdy and Schmidt (1996), the Mayans cultivated cocoa for 2000–4000 years before Spanish contact. *Theobroma cacao*, now its official botanical name, was named in 1737 by the Swedish botanist Carolus Linnaeus from the Greek word "ambrosia," which refers to the tree's

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mythical background, literally meaning "cocoa, food of the gods" (Alvim 1984).

Globally, cocoa is grown on over 11.5 million ha of land between 10°N and 10°S of the equator, with Africa cultivating the majority (7.9 m ha) of the reported land area and Oceania the least (0.12 m ha) (Fig. 1) (FAOSTAT, 2023). Cocoa is a perennial tree crop that can grow to a height of 5–15 m (Predan et al. 2019) in areas with year-round rainfall, preferably between 1500 and 2500 mm per year, and high humidity, often 70–100% (Afoakwa 2016). Annual rainfall over 2500 mm has been linked to an increase in the incidence of fungal diseases such as black pod and vascular streak dieback, as well as heavy leaching of soil nutrients (van Vliet and Giller 2017). A study in Cameroon found a higher (65–70%) incidence of *Phytophthora* pod rot disease when rainfall exceeded 2200 mm (Ndoumbè-Nkeng et al. 2009). The crop also prefers temperatures ranging from 18 °C to 32 °C, varied soil conditions, and a dry season of no more than three months. The cocoa tree produces fruits, called pods along the trunk and branches. The pods are oval, 12 to 30 cm long, and contain about 30–40 seeds, also known as cocoa beans (Lima et al. 2011). The seeds are also oval-shaped, measuring about 2 cm long and 1 cm wide, and are embedded in a sweet mucilaginous pulp that accounts for about 40% of the fresh weight of the bean (Predan et al. 2019). The beans are made up of an embryo and two cotyledons (86–90% of the dry weight of the seed) (Santander et al. 2019).

Traditionally, cocoa is categorized into three (3) broad groups namely Forastero, Criollo, and Trinitario (Bhattacharjee and Akroda, 2018). A fourth type, Nacional, grows in Ecuador and is thought to have originated from the Amazonian area of Ecuador (Afoakwa 2014; Bhattacharjee and Akroda, 2018). These cocoa groups differ in chemical composition, pod and bean appearance, pest and disease resistance, flavour characteristics, and textural and organoleptic properties (Afoakwa 2016).

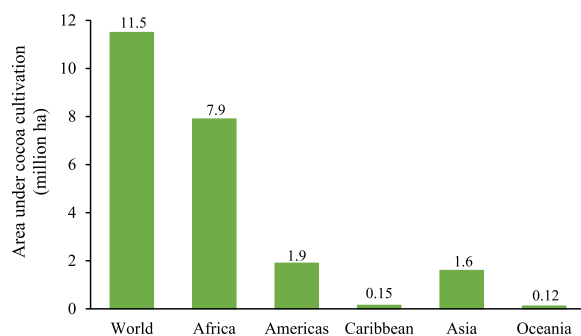


Fig. 1 Total land under cocoa cultivation in the producing regions in 2021. Source: Based on data from FAOSTAT, 2023

Within these groups are several varieties. Forastero cocoa, a higher-yielding and more robust variety, yields bitter-tasting beans that account for roughly 80% of the cocoa beans grown globally (Tridge 2021), while the Criollo produces a higher-quality aromatic and mildly bitter bean, accounting for less than 5% of total global production (Tridge 2021; Phayanak, 2023). The Trinitario cocoa, a cross between the Forastero and Criollo, accounts for approximately 15% of global production (Tridge 2021; Phayanak, 2023), whereas Nacional, on the other hand, is a rare Ecuadorian cocoa, accounting for about 2% of global production (Bermudez et al. 2022). The Nacional cocoa, which produces beans with a more floral aroma and little bitterness (Bar and Cocoa 2023), was wiped out by a plant disease epidemic that swept through Ecuador in 1920 (Bermudez et al. 2022). Forastero cocoa is commonly referred to as "bulk cocoa", whereas Criollo, Trinitario, and Nacional cocoa are classified as "fine flavour cocoa (FFC)" on the global cocoa market.

A study by Motamayor et al. (2008) proposed a new classification of cocoa germplasm into 10 major clusters: Marañon, Curaray, Criollo, Iquitos, Nanay, Contamana, Amelonado, Purús, Nacional, and Guiana. According to the authors, this new classification reflects the genetic diversity of cocoa available to breeders more accurately than the traditional classifications of Criollo, Forastero, and Trinitario. Other scientific studies have been conducted to assess cocoa genetic diversity since 2008 (Ji et al. 2013; Santos et al. 2015). According to Bhattacharjee and Akroda (2018), the majority of these studies are limited to a few geographical locations and three or four cocoa groups (Criollos, Forasteros, Trinitarios, and Nacional or Cacao Nacional Boliviano types). A comprehensive study combining molecular and modelling approaches is required to understand cocoa's genetic diversity and population structure.

Cocoa is consumed by millions of people worldwide in the form of chocolate, beverages, and other cocoa-related products, as it is the primary raw material used in the production of chocolate and other cocoa products (Kongor and Dimas, 2023). Global demand for cocoa is expected to rise as the global volume of sales of chocolate confectionery products rises by 1% from 7.54 million tons in 2022 to 7.61 million tons in 2023, with North America and Pacific Asia contributing the most to the increase in demand for chocolate confectionery products over the period (ICCO, 2023a). On the other hand, the cocoa bean supply for the 2022–2023 season lags behind 2021–2022 (ICCO, 2023b). Data on crop sizes in West Africa's main cocoa-producing countries suggest that, compared to the 2021–2022 cocoa year, the 2022–2023 cocoa season will have a supply deficit due to a decrease in production (ICCO, 2023b). The supply deficit has been

attributed to a growing number of uncertainties including the current global macroeconomic outlook, the effect of climate change, diseases, and the weather variation in West Africa (Myers 2023).

Cocoa production in the major producing countries is typically carried out by smallholder farmers on a low-input basis, often relying on nature such as rainfall. There are no prudent pest and disease management systems in the cocoa fields, as well as adequate soil fertility management, which leads to nutrient depletion in the cocoa fields (Akrofi et al. 2015; van Vliet and Giller 2017). Under the current cocoa production system, cocoa farmers in most of the major producing countries have low productivity and poor livelihoods, forcing them to convert forests into farmlands, which has negative effects on the environment, climate, and biodiversity (Kongor and

Dimas, 2023). This review examines the current major challenges in global cocoa production as well as the consequences of cocoa production on the environment, climate, and biodiversity. Increased understanding of current challenges, as well as practical solutions for sustainably addressing these challenges, would have significant economic and industrial implications.

Importance of cocoa

Cocoa is a major agricultural commodity that is traded and consumed around the world. The crop is important because it provides economic benefits to producing countries, smallholder farmers cultivating the crop, and the confectionery industry by providing key raw materials. (Kongor and Dimas, 2023). Global annual cocoa production for the 2021–2022 season was estimated at 4.82 million tons, with Africa contributing 74.5%, while Asia and Oceania accounted for 5.5% (Fig. 2) (ICCO, 2023c). Cocoa beans were reported to be the 423rd most traded product in the world in 2022, with a total trade value of \$8.29 billion (Observatory of Economic Complexity [OEC], 2024). Cocoa bean exports decreased by 20.3% between 2021 and 2022, from \$10.4 billion to \$8.29 billion (OEC, 2024). With an export value of US \$3.33 billion in 2022, Cote d’Ivoire was the largest producer and exporter of cocoa beans, followed by Ghana (US \$1.08 billion), Ecuador (US \$937 million), Nigeria (US \$489 million), and Cameroon (US \$450 million) (Fig. 3) (OEC, 2024). Cocoa plays an important role in the economies of most producing countries. It is a significant source of income for governments in cocoa-producing countries,

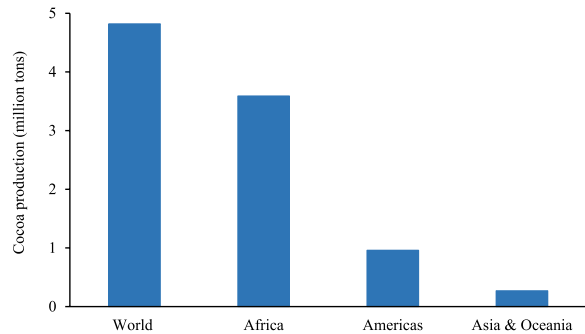


Fig. 2 Cocoa production (million tons) in the cocoa-growing regions. Source: Based on data from the ICCO, 2023c

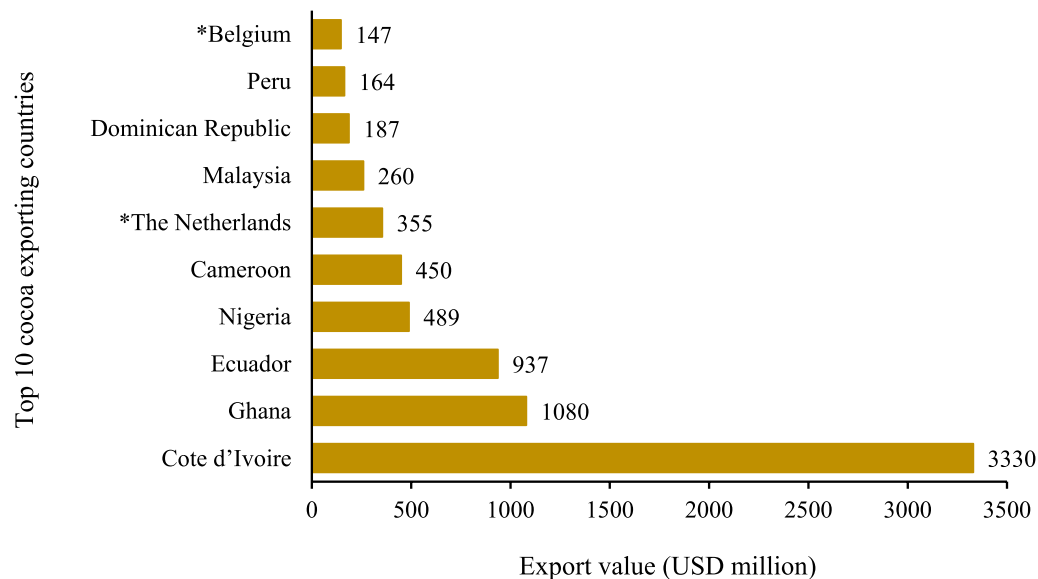


Fig. 3 Major cocoa exporting countries in the world. Source: Based on data obtained from Observatory of Economic Complexity [OEC], 2024) *The Netherlands and Belgium are not cocoa-producing countries but re-export imported cocoa beans

the leading foreign exchange earner, and contributes significantly to GDP in the majority of these producing countries (Kehinde, 2022; Oyekale 2022). Cocoa is the fourth-largest agricultural export commodity in Indonesia after palm oil, rubber, and coconut (Indonesia-Investment, 2024), third in Papua New Guinea after palm oil and coffee, accounting for approximately 14% of national agricultural export revenue (Swan et al. 2024), and maintaining a 2.5% GDP growth rate in Ecuador (Salazar et al. 2023).

Cocoa also provides direct and indirect employment, income, and a means of subsistence for smallholder farmers and other actors in the value chain (Kehinde and Ogundeji, 2022; Wijayati and Haqqi 2022; Swan et al. 2024). Cocoa helps to build wealth and provide food, jobs, and resources for eradicating poverty (Osei-Gyabaa et al. 2023). Global cocoa production relies almost entirely on 5–6 million smallholder farmers (Bermudez et al. 2022). The fact that more than 800,000 farmers in Ghana, approximately 300,000 farmers in Nigeria, 600,000 farmers in Cameroon, and more than 1 million farmers in Cote d'Ivoire are primarily engaged in cocoa cultivation (United Nations Development Programme [UNDP], 2021; Oyekale 2022) demonstrates the importance of cocoa to the rural economies of the producing countries. Cocoa provides between 60 and 90% of rural farmers' household incomes (Voora et al. 2019), which is used to care for the entire family, buy food, and pay school fees (Bymolt et al. 2018). Kongor and Dimas (2023) noted that several activities in cocoa production, such as weed control, pruning, fertilizer application, pests and disease control, harvesting, pod breaking, marketing functions and value addition represent a viable source of employment and income generation leading to economic empowerment in rural households.

Cocoa contributes to economic growth in developed countries, particularly in Europe and the United States, by providing critical industrial raw materials for the confectionery industry. Although the COVID-19 pandemic had an impact on the global chocolate confectionery market because many countries regarded chocolate confectionery as a non-essential item during the pandemic (Indranil et al. 2022), the market recorded a revenue of approximately 0.99 trillion US dollars in 2021 (Statista, 2022). The market is also projected to increase to 1.33 trillion dollars in 2027, according to Statista Digital Market Outlook estimates (Statista, 2022).

Aside from its economic importance, the consumption of cocoa and other cocoa-related products has received increased global attention in recent years, owing primarily to its bioactive compounds, which have been shown to benefit human health (Samanta et al. 2022; Kongor et al. 2023a, b). The major bioactive compounds found

in cocoa include polyphenols (mainly flavan-3-ols, with minor amounts of anthocyanins and flavanols) and methylxanthines (mainly theobromine [3, 7-dimethylxanthine] and caffeine [1, 3, 7-trimethylxanthine]) (Febrianto et al. 2022). Consumption of cocoa and cocoa-related products has been linked to improved cardiovascular health (Araujo et al. 2016; Kongor et al. 2023a, b). The best-established health benefit of cocoa flavanols, according to Goya et al. (2022), is their positive effect on cardiovascular function. Cocoa flavanols contribute to normal blood flow by maintaining normal blood pressure (Baynham et al. 2021) and endothelium-dependent vasodilation (Navarrete-Yaez et al. 2021). The methylxanthines in cocoa have important biological activities that have been linked to a variety of health benefits, including the prevention of respiratory and cardiovascular diseases, cancer, obesity, diabetes, and neurodegenerative diseases (Monteiro et al. 2019; Cadoná et al. 2022).

Structure of the cocoa industry

The structure of the cocoa industry can be broadly categorized into upstream and downstream sectors. The upstream sector is made up of the production and marketing of cocoa beans and is dominated by cocoa-producing countries, whereas the downstream sector consists of cocoa bean processing activities to obtain semi-finished and finished products. Downstream activities are carried out in both producing and importing countries. The cocoa market is primarily driven by rising global demand for cocoa-related products like chocolates, chocolate coatings, and cosmetics (Beg et al. 2017). Although cocoa originated in South America's tropical humid forests (Caligiani et al. 2016), it is now grown in the Caribbean, Asia, Africa, and the Pacific, including Papua New Guinea, Fiji, the Solomon Islands, Samoa, and Hawaii (Hebbar et al. 2011). The global cocoa value chain begins with cocoa farmers in the producing countries. Farmers cultivate farmland to produce cocoa pods. The mature and ripe pods are then harvested and processed to produce dried cocoa beans. It is estimated that 5–6 million smallholder farmers produce 90% of the world's cocoa, typically on farmlands ranging from 2–5 ha with low inputs and technology, while only 5% of global production comes from plantations that are larger than 40 ha (CocoaNet 2022; Tridge 2021). Upstream cocoa processing includes harvesting, pod storage, pod breaking, fermentation, drying, and bagging of dried fermented beans. The beans are then sold to certified buyers, who resell them either domestically or globally for downstream processing.

According to ICCO data, Africa produced over 7.6 million tons of the global 10 million tons of cocoa beans from the 2020–2021 to 2021–2022 season, accounting

for 76% of global production during the period (ICCO, 2023c). Côte d'Ivoire is the world's largest cocoa producer, producing approximately 2.2 million tons of cocoa beans per year, while Ghana is the world's second-largest cocoa producer, accounting for about 20% of global cocoa production (ICCO, 2023c). Ghana produced a little over 1 million tons of cocoa beans in 2020–2021, the most ever produced by the country (Osei-Gyabaah et al. 2023), which has been projected to decrease in the 2021–2022 season due to the annual fluctuations in the global price of cocoa beans (Oluwole 2022). However, in terms of quality cocoa beans, Ghana ranks first as the world's leading producer and exporter of high-quality premium “bulk cocoa”, and its beans are mostly used as a standard reference against which other cocoa beans are measured (Teye 2022). Other major cocoa producers in Africa include Nigeria, Cameroon, and Uganda. Cameroon's cocoa production has increased over the last 20 years, rising from 122,600 tons in 2000 to 290,000 tons in 2020, owing largely to the expansion of the cocoa production areas (Suh and Molua 2022). Furthermore, cocoa production in Uganda has steadily increased in recent years, with over 30,000 tons of cocoa beans produced each year, according to Uganda Export Promotion Board data (Oluwole 2022). Amelonado is the most common *Forastero* cocoa variety grown in West Africa, though crosses between *Trinitario* and Amelonado are also widely distributed to farmers in the region (Pokou et al. 2009; Padi et al. 2015).

Cocoa production in the Americas has increased overall, from 933,000 tons (17.8%) in the 2020–2021 season to an estimated 963,000 tons (20.0%) in the 2021–2022 season (ICCO, 2023c). In the 2022–2023 season, production is expected to reach 988,000 tons, accounting for 19.8% of global output (ICCO, 2023c). Ecuador, Brazil, Peru, Colombia, Dominican Republic, Nicaragua, and Mexico are the major producing countries (Hütz-Adams et al. 2022). Ecuador is the most important producer in the Americas and the world's third-largest producer of cocoa beans, producing 365,000 tons (approximately 7.6%) of global cocoa beans in 2021–2022 and a projected 400,000 tons (8.0%) in 2022–2023 (ICCO, 2023c). However, the production volume in Ecuador is still less than half of Ghana's and one-fifth that of Côte d'Ivoire (Hütz-Adams et al. 2022). According to Cortez et al. (2024), although Peru is not a major producer like Brazil or Ecuador, its cocoa is highly valued in the European market. Latin America is the world's leading producer and exporter of fine-flavour cocoa, accounting for 90% of total fine-flavour cocoa exports, with Ecuador, the Dominican Republic, and Peru currently being the top three fine-flavour cocoa exporters (ICCO, 2023d).

Indonesia is the leading cocoa producer in Asia and Oceania, followed by Papua New Guinea. For many years, Indonesia has been the region's leading cocoa producer, and it was once the world's third-largest cocoa bean exporter after Ghana and Côte d'Ivoire (ASEAN Today 2019). Between the 2005–2006 and 2009–2010 seasons, Indonesia produced over 500,000 tons of cocoa beans per year, accounting for approximately 15% of global production (Hütz-Adams et al. 2022). However, since 2010–2011, production has declined steadily by two-thirds (Hütz-Adams et al. 2022). The main reasons for the decline in productivity since 2011 have been attributed to ageing cocoa trees, high pest and disease losses, soil fertility exhaustion, and a failure to implement Good Agricultural Practices (GAP), as well as the failure of poorly funded provincial extension services to reach many farmers (Moriarty et al. 2014; Arsyad et al. 2019). According to the International Cocoa Organization's (ICCO) most recent production data, Indonesia produced 170,000 tons (3.2%) of cocoa beans in the 2020–2021 season and is expected to produce 180,000 tons (3.6%) in the 2022–2023 season (ICCO, 2023c). Malaysia, India, and Fiji are also significant producers in Asia and Oceania.

It is one thing to grow cocoa trees; it is quite another to process cocoa beans into semi-finished and finished products. Although West Africa produces the majority of cocoa, Europe and the United States dominate the downstream cocoa sector. Beg et al. (2017) noted that the Dutch were the first to trade cocoa beans and controlled the global cocoa trade until the eighteenth century. Europe has the highest industrial demand for cocoa beans and is the world's largest importer; accounting for 56% of global imports, while North and Latin America account for approximately 17% and Asia accounts for 26% (Centre for the Promotion of Imports [CBI], 2022). In 2021, Europe's total cocoa bean imports exceeded 2.2 million tons, with nearly 79% sourced directly from producing countries, amounting to nearly 1.8 million tons (CBI, 2022). With an import value of US \$1.68 billion in 2022, the Netherlands was the world's largest importer of cocoa beans, followed by Malaysia (US \$1.02 billion), the US (US \$874 million), Germany (US \$681 million), and Belgium (US \$678 million) (OECD, 2024).

Global cocoa bean grinding was estimated at 4.99 million tons for the 2021–2022 season, with Europe accounting for 35.5%, Asia and Oceania (23.1%), Africa (22.7%), and the Americas (18.7%) (Fig. 4) (ICCO, 2023e). However, Europe's share of global grinding is said to have decreased slightly in recent years as cocoa grinding in producing countries has increased. Cocoa grinding at origin, for example, was estimated at 2.33 million tons in 2021–2022, accounting for 46.8% of all grinding activities worldwide (ICCO, 2023e). It is forecasted to reach 2.41

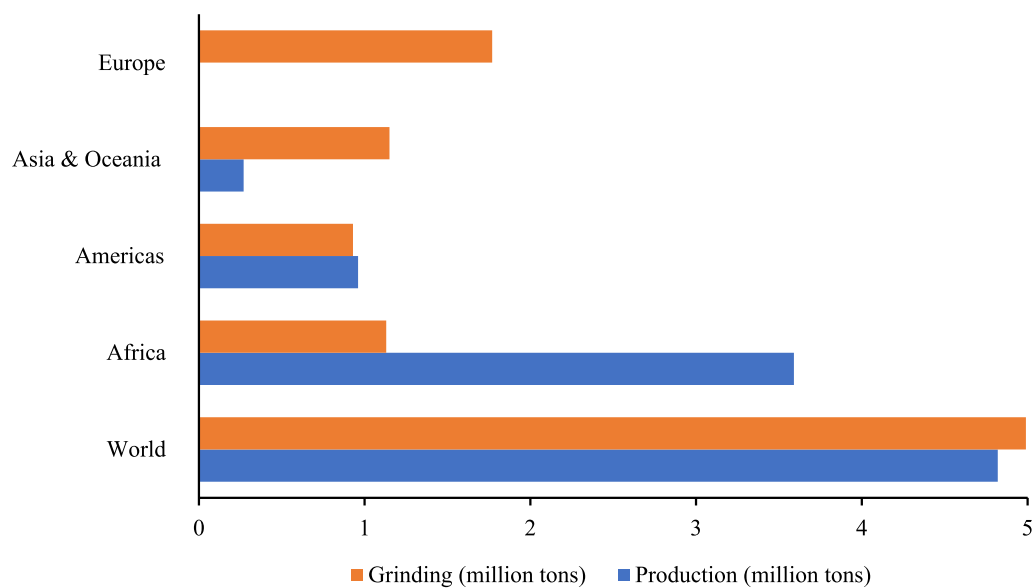


Fig. 4 Global cocoa production (million ha) and grinding (million tons). Source: Based on data from the ICCO, [2023c](#); [2023e](#)

million tons (47.5%) in the 2022–2023 season (ICCO, [2023e](#)). Large multinationals such as Cargill, Olam, and Barry Callebaut have used grinding at origin to reduce production costs while also targeting regional markets. Although Europe accounts for the majority of global grindings, Cote d'Ivoire was the world's largest cocoa grinder in the 2020–2021 and 2021–2022 seasons, with 620,000 tons and 710,000 tons, respectively, followed by the Netherlands, with 610,000 tons in both seasons (ICCO, [2023e](#)). Regardless, Africa still exports the majority of its beans.

According to available data, the majority of cocoa beans produced in the Americas are processed within the region (Fig. 4). Ecuador is by far the largest cocoa producer, but it grinds less than 10% of its crop domestically, whereas Brazil has become the only major cocoa-producing country that is a net cocoa importer (Hütz-Adams et al. [2022](#)). The Brazilian market consumes so much chocolate that domestic production is insufficient to meet demand. Peru and Colombia both have sizable local grinding industries and chocolate consumption. Hütz-Adams et al. ([2022](#)) reported that more than 90% of the Dominican Republic's output is exported. Nicaragua is also heavily reliant on exports, but the majority of these go to neighbouring countries. Mexico is distinct from the others in that it imports far more cocoa than it produces. Asia and Oceania region have very different characteristics. Grinding activities exceed production volumes, suggesting high volumes of imports (Fig. 4) to meet rising demand and cocoa consumption in the region. Malaysia and Indonesia are ranked third and fifth in terms of

cocoa imports in 2021, with import values of US \$1.2 billion and US \$616.9 million, respectively (Tridge, [2023](#)).

The environmental context

Cocoa is a bittersweet crop in that, while it serves as a major cash crop for the economies of producing countries and provides critical industrial raw materials for the chocolate confectionery industry, it also contributes significantly to environmental destruction. Cocoa has been responsible for more than 60% of agri-commodity-driven deforestation in cocoa-producing areas in Cote d'Ivoire, Ghana, and Cameroon since 2000 (Bermudez et al. [2022](#); Kouassi et al. [2023](#); Parra-Paitan et al. [2024](#)). According to Parra-Paitan et al. ([2024](#)), more than 95% of the cocoa produced in Cote d'Ivoire, Ghana, and Cameroon is grown in areas where cocoa is the dominant crop, accounting for more than half of deforestation caused by agri-commodities. Other tree crops, including rubber, coffee, and oil palm, have also been linked to environmental destruction (Dow Goldman et al. [2020](#); Teng et al. [2020](#); Poncet et al. [2024](#); Chiriaco et al. [2024](#)). Agri-commodity deforestation in cocoa-producing areas in South America and Indonesia is primarily driven by pasture for livestock feed and oil palm, respectively (Teng et al. [2020](#); Chiriaco et al. [2024](#); Parra-Paitan et al. [2024](#)). Cocoa appears to be the least responsible for deforestation in South America, with only ~3%, 5%, and 30% of volumes in Colombia, Peru, and Ecuador, respectively, accounting for more than 50% of deforestation (Parra-Paitan et al. [2024](#)). Parra-Paitan et al. ([2024](#)) also observed that cocoa-driven deforestation always coexists with other

commodities that drive deforestation, even in cocoa landscapes where it is the dominant driver. Robusta and Arabica coffee, for instance, grown in cocoa landscapes, contribute significantly to deforestation.

The environmental impacts of the cocoa industry are felt throughout the value chain, beginning with the production of cocoa beans in producing countries and continuing through the industrial processing of beans into semi-finished and confectionery products. Some of the environmental consequences of cocoa are obvious, such as loss of forest and biodiversity, while others are less obvious, such as greenhouse gas (GHG) emissions as a result of deforestation and the fossil fuels used to transport the beans to the processing plant (Vervuurt et al. 2022). Cocoa cultivation also adversely affects the environment through soil degradation and contamination caused by the use of chemicals to control weeds, pests, and diseases and improve soil fertility (Perez-Neira et al. 2020). Several researchers believe that cocoa cultivation has a greater environmental impact than the downstream activities of cocoa bean processing and chocolate manufacturing (Vervuurt et al. 2022; Bermudez et al. 2022; Kouassi et al. 2023; Kalischek et al. 2023). A study in Cote d'Ivoire by Vervuurt et al. (2022) for instance discovered that, on average, the production of 1 kg of cocoa beans produced 1.47 kg of CO₂eq. Boakye-Yiadom and his co-workers (2021), on the other hand, are of a different mind. Their research in Ghana discovered that the chocolate manufacturing phase was generally more damaging to the environment than cocoa cultivation due to high GHG emissions from milk and sugar production, both of which are used in chocolate production (Boakye-Yiadom et al. 2021).

Cocoa farming has played a substantial role in the deforestation of West African Upper Guinean forests (a biodiversity hotspot) that has occurred in waves over the twentieth and twenty-first centuries (Kalischek et al. 2023). According to Fountain and Huetz-Adams (2020), cocoa production continues to put pressure on rainforests in the Congo Basin, the Amazon Basin, Colombia, and Indonesia. The Amazon region, which is one of the world's biodiversity hotspots, is currently under stress due to increased agricultural activity, the spread of monoculture, and the excessive use of synthetic chemical fertilizers and pesticides (Vasco et al. 2021; Caicedo-Vargas et al. 2022). However, because of the smaller areas used in Latin America for cocoa cultivation, the damage has been less severe than in West Africa (Hütz-Adams et al. 2022). Some Southeast Asian areas, such as North Sumatra, East Kalimantan, Sulawesi, East Sepik, Madang in Papua New Guinea, and Sarawak in Malaysia, are expected to become more attractive for cocoa expansion, potentially increasing the risk of deforestation (Parra-Paitan et al.

2024). According to Meyfroidt et al. (2018), this can happen either directly through forest encroachment or indirectly through the displacement of other land uses elsewhere.

To mitigate the negative environmental impact of cocoa production, several studies have proposed cocoa agroforestry as a viable alternative to cocoa monoculture (Utomo et al. 2016; Asitoakor et al. 2022a; Hawkins et al. 2024). Cocoa agroforestry is a planned, intentional system of growing cocoa alongside other trees and crops over time and space (Sanial et al. 2020). It is common in Central and South America, but also in some African countries such as Cameroon, Nigeria, and Côte d'Ivoire (Jagoret et al. 2011; Oke and Odebiyi 2007), where farmers manage their cocoa farms by preserving some of the forest species present when the cocoa farm was established while eliminating others and introducing fruit species into the system (Mattalia et al. 2022). In cocoa monoculture systems, on the other hand, cocoa is grown under full sun, with little or no shade, but with high inputs (fertilizers and agrochemicals). This system is widely used in Côte d'Ivoire, Ghana, Ecuador, Peru, Malaysia, and Indonesia (Belsky and Siebert 2003; Schneider et al. 2017). Cocoa agroforestry is reported to provide a variety of ecological benefits, including biodiversity conservation for flora and fauna, carbon sequestration, preserving and strengthening soil moisture and fertility, pest control, and microclimatic controls such as stimulating rainfall, among many others (Utomo et al. 2016; Sanial et al. 2020; Marconi and Armengot 2020; Kouassi et al. 2021). Blaser-Hart et al. (2021) emphasized that agroforestry is a climate-smart strategy used by agricultural stakeholders to combat climate change and improve agricultural production sustainability.

In contrast to the aforementioned positive contributions of cocoa agroforestry systems, some studies (Ahensorah et al. 1987; Armengot et al. 2016) have reported higher cocoa yields under cocoa monoculture systems but with high inputs (fertilizers and agrochemicals). Grant et al. (2022) identified excessive shade cover in cocoa agroforestry farms, poor canopy architecture of shade trees, inconsistent growth requirements of cocoa trees, the impact of intercropping on cocoa agroforestry, the spread of pests and diseases by some shade trees, and competition for soil nutrients as limiting factors to cocoa agroforestry systems. Due to perceived competition for light, water, and nutrients, some farmers have removed shade trees from their farms in response to these contrasting results. Niether et al. (2020) also observed that, despite all of the benefits of cocoa agroforestry systems, lower cocoa yields may still be one of the most important factors impeding broader adoption of diversified production systems, and they recommended additional

research focused on increasing cocoa yields in agroforestry systems through shade-tolerant variety breeding or adapted management practices to increase pollination rates. Some earlier studies, on the other hand, have demonstrated that the negative interactions between cocoa and shade trees in agroforestry systems can be mitigated by carefully selecting shade tree species and implementing management practices such as pruning or thinning (Blaser-Hart et al. 2021; Asitoakor et al. 2022a). Utomo et al. (2016) discovered that not all trees are beneficial in cocoa agroforestry and that the cocoa-coconut agroforestry system performed better in Indonesia than other cocoa monoculture and cocoa-rubber agroforestry systems across all environmental impact categories.

Pests and diseases

The cocoa tree is susceptible to various pests and diseases that reduce tree yield and constitute one of the biggest concerns regarding cocoa production (Table 1). Marelli et al. (2019) noted that diseases alone account for approximately 38% of global cocoa losses. Depending on the variety of cocoa and region, insect pests and diseases have been identified to attack different parts of cocoa trees and the pods, resulting in economic loss (Oyenpemi et al. 2023). Most of these diseases are caused by fungi and viruses. Five (5) major diseases, namely black pod (*Phytophthora* pod rot), cocoa swollen shoot virus (CSSV), frosty pod rot, vascular streak dieback, and witch's broom, are known to affect the crop (Fig. 5). Black pod disease and cocoa swollen shoot virus disease are the most common diseases affecting cocoa in West African countries (Akrofi, 2015; Asitoakor et al. 2022b; Oyenpemi et al. 2023). Witches broom disease (caused by *Moniliophthora perniciosa*) and frosty pod rot (caused by *Moniliophthora roreri*) are major causes of production losses in the Americas, whereas vascular streak dieback (caused by *Oncobasidium theobromae*) is the major

disease in most Asian producing countries (Sousa Filho et al. 2021; Asman et al. 2021; Bryceson et al. 2023). In West African-producing countries, the most common pests are mirids, also known as capsids (*Distantiella theobroma* [Distant], *Sahlbergella singularis* [Haglund], and *Helopeltis spp.*), and cocoa shield bugs (*Bathycoelia thalassina*), whereas in most Asian producing countries, the cocoa pod borer (*Conopomorpha cramerella*) is the main pest responsible for production losses (Fig. 6).

Black pod disease

The black pod disease (a fungal disease) is regarded as the most devastating of all cocoa diseases, causing significant economic losses of more than 30% of global cocoa production, while individual farms may suffer harvest losses ranging from 30 to 90% (Adeniyi and Asogwa 2023), making it the most significant production constraint. The disease was once thought to be caused by a single species, *Phytophthora palmivora*, but increased knowledge has shown that this is not the case, and each continent has a complex of *Phytophthora* species that can cause black pod disease in cocoa (Adeniyi 2019). Currently, more than seven species of the genus *Phytophthora* have been identified as black pod disease agents worldwide, including *P. palmivora*, *P. megakarya*, *P. citrophthora*, *P. megasperma*, *P. arecae*, *P. heveae*, *P. capsici*, *P. tropicalis*, and *P. cacaicola* (Aragaki and Uchida, 2001; Kroon et al. 2012; Decloquement et al. 2021; Torres-de la Cruz et al. 2023). The most prevalent is *Phytophthora palmivora*, which is present in most cocoa-growing countries worldwide and causes 20–30% yield losses and 10% tree deaths annually (Adeniyi 2019). *P. megakarya* is the most destructive pathogen in Central and West Africa, and it is the most aggressive pathogen among the black pod pathogens (Akrofi, 2015; Asitoakor et al. 2022b; Oyenpemi et al. 2023). *P. capsici* is common in Central and South America, where it causes significant losses in favourable

Table 1 Dominant cocoa diseases and pests in the cocoa-growing regions

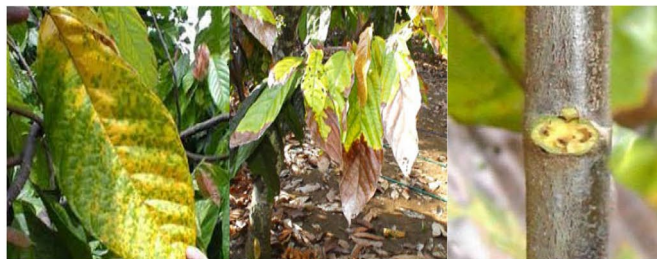
Disease	Causal agent	Major affected region
Black pod	<i>Phytophthora spp.</i>	West Africa, the Americas, Asia, the Caribbean
Witches' broom	<i>Moniliophthora perniciosa</i>	Caribbean, South America
Cocoa swollen shoot virus disease	Cocoa swollen shoot virus	West Africa
Vascular streak dieback	<i>Oncobasidium theobromae</i>	Melanesia, Southeast Asia
Frosty pod rot	<i>Moniliophthora roreri</i>	Central and South America, Caribbean
Pests		
Mirids (Cocoa capsids)	<i>Distantiella theobroma</i> <i>Sahlbergella singularis</i> , <i>Helopeltis spp.</i> <i>Monalonion spp.</i>	West Africa, West Africa, Southeast Asia, Latin America
Cocoa pod borer	<i>Conopomorpha cramerella</i> , <i>Carmenta theobromae</i> , <i>Carmenta forasemini</i>	Pacific Islands, South Asia, Southeast Asia, Latin America, Latin America
Cocoa shield bug	<i>Bathycoelia thalassina</i>	West Africa



A) Black pod disease-infested cocoa pod and causing the rotting of the beans



B) CSSVD showing as swollen shoot and various shades of chlorosis in mature leaves



C) Vascular streak dieback showing yellow leaf, yellowing of leaves on shoots, and leaf scar with discoloured vascular tissues



D) Witches broom showing external and internal rotting of pod



E) Cocoa pods affected by frosty pod rot

Fig. 5 Major cocoa diseases in the cocoa-growing regions. Sources: **A** and **B** Authors own pictures taken on the field; **C** Guest and Keane, 2007; **D** Ministry of Food Production, Trinidad and Tobago; **E** Cubillos, 2017

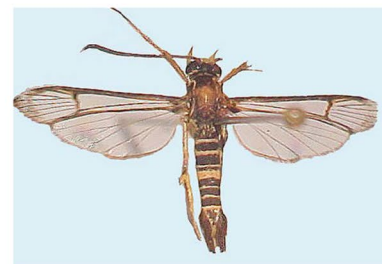
A) *Distantiella theobroma*B) *Sahlbergella singularis*C) *Helopeltis* spp.D) *Monalonion* spp.E) *Bathycoelia thalasina* (Cocoa shield bug)F) *Bathycoelia thalasina* on a cocoa podG) *Conopomorpha cramerella*H) *Carmenta theobromae*

Fig. 6 Major cocoa pests in the producing regions [A–D: major capsids in the different cocoa-growing regions; G and H are cocoa pod borers].

Sources: A–C & F: http://www.dropdata.org/cooa/icm_bkp.htm#vsd; D – www.invesa.com; E Authors own picture; G www.inaturalist.com; H Puchi, 2005

environments. In addition to *P. capsici*, and *P. palmivora*, two other *Phytophthora* species, *P. citrophthora* and *P. heveae*, have been linked to high levels of black pod disease in Brazil and Mexico (Luz et al. 2018; Adeniyi 2019; Lessa et al. 2020). *P. tropicalis* (Aragaki and Uchida,

2001), *P. cacaicola* (Decloquement et al. 2021), and *P. parasitica* (Torres-de la Cruz et al. 2023) have all been linked to black pod disease in the Americas.

The disease has the potential to reduce cocoa yield by 40–90% in West Africa (Kehinde and Tijani 2021), a

region that accounts for 60–70% of global production. This poses a significant risk to the cocoa industry and its producers. A production loss of between 20 and 25% in Mexico has been reported (Torres-de la Cruz et al. 2023). McMahon et al. (2015) noted that the incidence of black pod disease as well as vascular streak dieback (VSD), is driving Indonesian farmers to replace cocoa with other crops, particularly oil palm. The disease causes brown or black lesions on pods, resulting in premature pod rot and falling, reducing crop yield and quality (Adeniran et al. 2024). The disease can eventually spread throughout the pod, and once the husk is infected, *Phytophthora* invades the internal pod tissues, causing cocoa beans to discolour and shrivel (Acebo-Guerrero et al. 2012). The fungus spreads quickly on cocoa pods when there is too much rain and high humidity, insufficient sunlight, and temperatures below 21 °C. The pathogen spreads via infected pods, fallen fruits, and infected pod debris on the ground, making control difficult (Adeniran et al. 2024).

Currently, black pod disease control measures combine chemical and cultural methods. Fungicides, such as copper-based formulations, are widely used to prevent and treat fungal infections on pods (Adeniran et al. 2024). However, frequent fungicide application can be costly, environmentally harmful, and result in fungal resistance (Ndoumbe-Nkeng et al. 2004). Cultural practices such as removing infested pods and controlling weeds have been shown to reduce disease incidence and severity (Soberanis et al. 1999; Acebo-Guerrero et al. 2012; Adeniran et al. 2024). Removing infested pods eliminates potential inoculum sources and lowers humidity around pods, resulting in a less favourable environment for fungal growth (Adeniran et al. 2024). In Cameroon, Ndoumbe-Nkeng et al. (2004) found that weekly removal of diseased pods from farmers' plots reduced the black pod rate by 22–31% in the first year and by 9–11% in the second year, compared to a plot with no preventive control measures. Earlier studies in Peru by Soberanis et al. (1999) found that weekly phytosanitary pod removal reduced black pod incidence by 35–66%.

Genetic improvement of cocoa varieties for resistance to black pod disease has also been carried out (Nyassé et al. 2007; McMahon et al. 2015; Ofori et al. 2023). Despite extensive research into cocoa genetic modification and numerous breakthroughs, fully resistant cocoa genotypes are currently unavailable (Nyassé et al. 2007; Acebo-Guerrero et al. 2012). There is also an increasing interest in the use of microorganisms as an alternative for plant disease management (Deberdt et al. 2008; Hanada et al. 2009). Several studies have used *Trichoderma* species to control black pods in vitro and in vivo (Hanada et al. 2009; Bailey et al. 2009), as well as *Bacillus amyloliquefaciens*, *Aspergillus* sp., and *Penicillium*

sp. in vitro and the field (Larbi-Koranteng et al. 2020). However, little has been done to assess the true economic benefit of biocontrol strategies versus chemical control for cocoa yield. There is also little molecular knowledge about the genes and metabolites that are formed during interactions between antagonists, plants, and pathogens. According to Acebo-Guerrero et al. (2012), the next step in characterizing cocoa defence mechanisms should be to identify the genes involved in such interactions. A comprehensive investigation of an integrated control strategy that effectively combines all of these methods (biological, chemical, genetic control, and appropriate cultural methods) while considering their economic viability and environmental impact is needed for the sustainable management of black pod disease.

Cocoa swollen shoot virus disease

The cocoa swollen shoot virus disease (CSSVD) is a viral disease that is a major problem in West Africa. The CSSVD's economic significance stems from its incapacitating and destructive effect on the cocoa tree, which can occur within a short period. It is regarded as the most serious cocoa viral disease in West Africa because it can cause 15–50% yield loss if severe strains are involved in infections (Muller et al. 2018). CSSVD has been reported to cause 80% yield reductions in Ghana and may eventually kill cocoa trees within a few years of infection (Amon-Armah et al. 2021). According to estimates, the disease has destroyed approximately 300 million cocoa trees in Ghana (Andres et al. 2018; Amon-Armah et al. 2021), and the resulting economic problems have resulted in the mass exodus of many early cocoa-producing communities and towns, particularly in Ghana's Eastern Region, where the disease was first identified (Amon-Armah et al. 2021).

The virus affects all parts of the cocoa plant, causing different symptoms in infected plants' leaves, stems, roots, and pods. Red vein banding in young leaves is one of the symptoms, which may be followed by vein clearing or chlorosis along the veins (Ofori et al. 2022). Swellings in stems, roots, and chupons, as well as distortion in pod shape and size, with affected pods becoming almost round or spherical, may occur concurrently or after the leaf symptoms (Ofori et al. 2022). CSSV is semi-persistently transmitted on cocoa by several mealybug species (*Pseudococcidae*, *Homoptera*) (Ameyaw 2019). The vectors eat everything on the cocoa tree, including the flowers, cherelles, pods, and leaves. The ability of the mealybug species to transmit different strains of the virus varies. *Planococcoides njalensis* (Laing), *Planococcus citri* (Rossi), and *Ferrisia virgata* (Okll), which are prevalent in cocoa fields in Ghana and Cote d'Ivoire, are the most efficient virus transmitters. Control measures

include removing infected trees and adjacent trees and completely burning them. In addition to these control measures, research is being conducted on alternative preventive measures such as resistance breeding, the use of mild strains for cross-protection, CSSV-immune crops as a barrier between farms, and mealybug vector control (Amon-Armah et al. 2021). In terms of CSSVD's impact on cocoa bean quality, there is no evidence that the disease has any negative impact on the quality of the cocoa tree's fruits or the quality of the beans after fermentation. This is because no comprehensive investigation has been conducted.

Vascular streak dieback disease

Vascular streak dieback (VSD) disease is a fungal disease caused by *Ceratobasidium theobromae*. It is one of the cocoa diseases responsible for decreasing cocoa production in Southeast Asia, as it kills many young trees and causes dieback on mature trees. VSD was first reported in Papua New Guinea in the late 1960s (Guest and Keane 2018), and later in Malaysia and Indonesia (Suhaida et al. 2021; Bryceson et al. 2023). Currently, the disease is only found in Southeast Asia and parts of Melanesia (Asman et al. 2021). Even though the disease was first reported in Papua New Guinea, it is now considered a minor disease due to the planting of more resistant cocoa genotypes (Asman et al. 2021). In Malaysia, the average yield loss due to VSD disease ranges between 10 and 15% (Suhaida et al. 2021). The pathogen that causes VSD is spread by wind at night and is thought to enter plant tissue through young leaves (Asman et al. 2021). Spores do not spread far, with very few infections occurring more than 80–100 m from the source of the inoculum (Guest and Keane 2018). Necrotic lesions and chlorosis on the leaf, as well as dark vascular discolouration when branches are split, are symptoms of the disease (Asman et al. 2021). When diseased leaves fall from symptomatic branches, the three vascular bundle traces visible on leaf scars become darkly discoloured.

Current VSD control strategies include sanitation pruning of diseased branches, which is sometimes considered impractical, particularly on infected young cocoa trees and seedlings; the use of VSD-resistant genotypes, which has been promoted in Papua New Guinea, Malaysia, and Indonesia (Asman et al. 2021); and the use of endophytic *Trichoderma asperellum* as a biological control measure (Rosmana et al. 2018). Asman et al. (2021) believe that controlling *C. theobromae* is challenging due to its vascular nature, making fungicides ineffective in the field. Asman et al. (2021) also discovered that VSD resistance is linked to scion (branch and leaf) characteristics rather than rootstock characteristics, and proposed that VSD resistance breeding should focus on scions.

Witches broom disease

Witches' broom is a fungal disease caused by *Moniliophthora perniciosa* (Lisboa et al. 2020). After black pod disease, it is considered the second most damaging disease to the cocoa tree economically (De Souza et al. 2018; Moretti-Almeida et al. 2019). The infection causes a slew of broom-like shoots to sprout, hence the name "Witches' Broom." Brooms are alive and green when they form, but they eventually die, leaving the characteristic dry witch's broom attached to the tree. The Amazon Rainforest region, where the cocoa tree is native, is the epicenter of the occurrence of witches' broom disease (Sousa Filho et al. 2021). The disease is now thought to be restricted to South America, as well as parts of Central America and the Caribbean (De Souza et al. 2018). Fortunately, it does not occur in Africa and Asia, which account for approximately 85% of global cocoa production (De Souza et al. 2018). However, the spread of witches' broom managed to affect cocoa production across a wide geographic strip, overcoming natural barriers and even affecting Caribbean islands, demonstrating that the disease's spread could reach other parts of the world where the disease's occurrence is still unknown (Sousa Filho et al. 2021).

In cocoa-producing countries in South America and the Caribbean where witches' broom has occurred on plantations, cocoa bean production has been reduced by 70% on average (Moretti-Almeida et al. 2019). The disease's spread poses a significant threat to maintaining cocoa production levels in affected regions, as the disease reduces production and productivity in cultivation areas. As previously stated, the occurrence of witches' broom can contribute to the abandonment of cocoa cultivation areas, ultimately leading to the substitution with other crops. This is a serious issue because millions of small-holder farmers rely on the crop for a living. Controlling witches' broom involves removing diseased plant parts, using copper-based fungicides, using several species of microorganisms (e.g., *Trichoderma stromaticum*) as bio-control agents, and developing disease-resistant varieties (De Souza et al. 2018). Among these control measures, De Souza et al. (2018) believe that an integrated approach is more effective than a single strategy in terms of disease management and sustainability.

Frosty pod rot disease

Frosty pod rot (FPR), also known as moniliasis, is a fungal disease caused by *Moniliophthora roreri*. It is the most devastating cocoa disease in Central and Latin American cocoa-producing areas (Jiménez et al. 2021). The disease is widespread, affecting Peru, Ecuador, Colombia, Venezuela, Panama, Costa Rica, Nicaragua, Honduras, Guatemala, Belize, and Mexico (Cuervo-Parra et al. 2011; Dorado Orea et al. 2017; Crisostomo-Panuera et al. 2024).

and is the cause of approximately 80% of annual crop losses (Dorado Orea et al. 2017). The disease was previously limited to Central and South America but has been confirmed in Jamaica since August 2016 (Johnson et al. 2017), and there is growing concern that it will soon spread to the Dominican Republic and other Caribbean islands (Cubillos, 2020). Brazil is, however, thought to be free of the disease but at high risk (Cubillos, 2020).

The disease is twice as destructive as the black pod disease, more dangerous, and more difficult to control than the witches' broom disease (Phillips-Mora and Wilkinson 2007). Current losses vary greatly, ranging from 10 to 100% (Phillips-Mora and Wilkinson 2007), and are influenced by factors such as the length of time the disease has been present in a site, the age of the cocoa plantation, crop and disease management, the presence of neighbouring affected plantations, and weather conditions (Phillips-Mora and Wilkinson 2007; Dorado Orea et al. 2017). Ecuador, for example, has lost more than 40% of its production, equivalent to USD20 million per year, and Mexico has been the most affected country since the disease's introduction in 2005 (Dorado Orea et al. 2017). In 2013, high temperatures affected 70% of production, causing damage to 50% of cocoa production in the states of Chiapas and Tabasco (Dorado Orea et al. 2017). However, in a global context, the current annual loss from FPR is small due to the disease's limited range of affected producing regions (Ploetz 2016), but the potential danger presented by the disease is enormous (Phillips-Mora and Wilkinson 2007).

FPR causative agent, *M. roreri*'s only infectious propagules are spores, which develop only on the pods of *Theobroma* and *Herrania* species (Phillips-Mora and Wilkinson 2007). The inoculum source for FPR is the infected pod, the source of spores (Bailey et al. 2018). Spore movement is affected by air movement and rain, and spore germination and infection require free moisture, a condition promoted by high humidity. Cocoa pods become infected at a young age (0–3 months) and are less susceptible as they mature. The symptoms of frosty pod rot appear only on the pods, and the external symptoms typically appear 3–8 weeks after the initial infection. The Caribbean Agricultural Health and Food Safety Agency (CAHFSA, 2022) describes the major symptoms of FPR in sequential order to include the development of bumps or abnormal swelling of the pods, premature or irregular ripening of pods, the development of dark brown 'wet' sunken irregular lesions, the hardening and discolouration of internal flesh and beans, beans are liquid or jelly internally, the development of a thick (2–3 mm) mass of white or cream spores on the lesion, and finally the mummification of the pod. The infected pods are also heavier than healthy pods of the same level of development.

Several strategies for controlling FPR have been investigated, including chemical, cultural, biological, and genetic control measures (Krauss et al. 2010; Cubillos, 2017; Cubillos 2020). According to Dorado Orea et al. (2017), the most effective way to control the FPR is through integrated control management, which combines all of these control measures. Cultural control measures include proper field drainage, adequate tree height (down to 4 m), shade, diseased pod removal, and fungicide applications (Bailey et al. 2018). Schroth et al. (2000) found that reducing shade from 70 to 50%, in combination with other practices, reduced FPR by 90%. Soberanis et al. (1999) discovered that weekly removal of diseased pods significantly reduced FPR when compared to pod removal every two weeks.

Moniliophthora spp. can also be controlled using antagonistic microorganisms (Krauss et al. 2010; Dorado Orea et al. 2017). Although research into biological control of FPR is in its early stages, Evans et al. (2003) investigated the suitability of numerous candidates for FPR control from *T. gileri* in Western Ecuador, particularly *Clonostachys spp.* and *Trichoderma spp.* In Colombia, a strain of *Trichoderma spp.* inhibited *M. roreri* growth by 95% (Dorado Orea et al. 2017), whereas in Mexico, *Trichoderma harzianum* strain VSL291 (Cuervo-Parra et al. 2011) and *Trichoderma viridescens* strain ITV43 (Cuervo-Parra et al. 2014) inhibited growth by 72.7% and 86.5%, respectively. Genetic breeding to create new FPR-resistant varieties is thought to be both cost-effective and environmentally friendly (Cubillos, 2017; Bailey et al. 2018). However, Dorado Orea et al. (2017) noted that resistance or tolerance to frosty pod rot has proven to be a rare feature, having only been found in 5 cocoa genotypes out of over 600 accessions evaluated in the Tropical Agricultural Research and Higher Education Center's (CATIE) Cocoa Genetic Improvement Program. Dorado Orea et al. (2017) proposed additional research into the genome sequencing of *M. roreri* to gain a more complete understanding of its evolution, biology, and the development of strategies to improve disease control.

Major pests

Cocoa production is also hampered by a plethora of insect pests, the majority of which are economically significant, resulting in a substantial reduction in tree growth and yield (Cilas and Bastide 2020; Adeniyi and Asogwa 2023). Mirids, also known as capsids, are the most common pests that attack cocoa in almost all cocoa-producing regions (Boateng et al. 2023). *Distantiella theobroma* (Distant) and *Sahlbergella singularis* (Haglund) are common capsids in West African countries, *Helopeltis spp.* are common in Southeast Asia, whereas *Monalonion spp.* are common in the Americas.

The cocoa shield bug (*Bathycoelia thalassina*) is another economically significant pest in West Africa. Capsids destroy foliage and young pods, resulting in significant production losses (Denkyirah et al. 2016). Capsid infestations can reduce cocoa production by up to 75% and lower cocoa bean quality (Nkamleu et al. 2007). Capsids are said to cause 25% crop losses in Ghana (Padi and Owusu 1998) and 30–40% crop losses in Côte d'Ivoire (N'Guessan et al. 2013). The cocoa pod borer is the major pest in the majority of Asian-producing countries and Latin America. The cocoa pod borer, *Conopomorpha cramerella* has hampered the development of cocoa production in most cocoa-producing countries in Asia (Cilas and Bastide 2020). Malaysia, for instance, which produced up to 247,000 tons in 1990, now produces only 1,000 tons, owing primarily to this pest (Arshad et al. 2015; Cilas and Bastide 2020). Although CPB was not the only reason for this decline, it was a significant constraint. The dominant cocoa pod borers in Latin America are the *Carmenta theobromae* and *Carmenta foraseminis*. *Carmenta foraseminis*, for example, is a rapidly spreading invasive species in Latin America. It has been known for several years in Colombia, Venezuela, and Brazil, and is now in Peru's cocoa-producing region of Alto Huayabamba. As a result of the damage to the cocoa pods, the pest can cause losses ranging from 30 to 70% (Sotomayor-Parian and Soto-Cordova, 2018).

Cadmium in cocoa

Cadmium (Cd) is a non-essential heavy metal that can be toxic to plants, animals, and humans at low levels (Gramlich et al. 2018; Maddela et al. 2020). The European Union (EU), the International Agency for Research on Cancer (IARC), and the United States Environmental Protection Agency (USEPA) have identified Cd as a Category 1B, Group 1, and Class B carcinogenic metal, respectively (Barraza et al. 2017). Consumption of food with high Cd content may result in various diseases such as renal tubular dysfunction, kidney stones, disruption of calcium metabolism, and skeletal, endocrine, reproductive, and respiratory defects (World Health Organization [WHO], 2019; Romero-Estévez et al. 2019). Cadmium occurs naturally in soils as a product of volcanic eruptions, vegetation burning, and soil evolution (Gramlich et al. 2018), and from anthropogenic sources such as mining and industrial activities, as well as agricultural practices such as the application of phosphate-based fertilizers, sewage sludge, and livestock manure (Maddela et al. 2020; Cordoba-Novoa et al. 2023). It is more mobile in soils and easily absorbed by plants, and its distribution and availability vary depending on the soil's physicochemical properties (e.g., texture, pH, organic matter composition,

and effective cation exchange capacity), microbial community, and climate conditions (Amari et al. 2017).

Cd has been detected in a variety of crops, including cocoa beans and cocoa-related products (Chavez et al. 2015; Gramlich et al. 2018), causing global attention and public concern. Market-based studies show that chocolate products have higher Cd concentrations than other foods, with mean values exceeding $100 \mu\text{g kg}^{-1}$ (European Food Safety Authority [EFSA], 2012). Again, the EFSA noted that chocolate confectionary products contributed an average of 4.3% of the European population's weekly Cd intake (EFSA, 2012). In response to public health concerns raised by high Cd levels in chocolate products, the European Union (EU) established maximum allowable limits for Cd in cocoa-based products imported into the EU, which took effect in January 2019 (Commission Regulation (EU), 2014). These limits are 0.1 mg kg^{-1} of Cd for chocolates with less than 30% dry cocoa solids, 0.3 mg kg^{-1} for chocolates with 30% to 50% dry cocoa solids, 0.8 mg kg^{-1} for chocolates with more than 50% dry cocoa solids, and 0.6 mg kg^{-1} for cocoa powder sold as a finished product to consumers. Cocoa beans from Latin America typically have higher Cd concentrations than those from West Africa (Cordoba-Novoa et al. 2023). Studies have reported average Cd levels of 0.94 mg kg^{-1} in Ecuadorian beans (Chavez et al. 2015), 0.21 mg kg^{-1} in Bolivia (Gramlich et al. 2017), over 0.8 mg kg^{-1} in 57% of cocoa bean samples from Peru (Arévalo-Gardini et al. 2017), and between 0.1 and 1.8 mg kg^{-1} across 6 geographical substrates in Honduras (Gramlich et al. 2018). Zarcinas et al. (2004) and Ramtahal et al. (2016) also found mean levels of 0.66 mg kg^{-1} with a maximum of 1.68 mg kg^{-1} and 0.35 and 3.82 mg kg^{-1} in cocoa beans grown in Malaysia and Trinidad and Tobago, respectively, indicating that the problem is not limited to the Americas.

Soil amendments and other cultural practices are among the strategies used to reduce Cd uptake in cocoa. Gramlich et al. (2018) found that soil pH, organic matter, and geology all influenced the availability of Cd to cocoa plants, implying that amendments could be effective in mitigating Cd contamination. Lime and biochar were found to be effective in reducing Cd uptake and translocation levels in cocoa under greenhouse and in vitro conditions in Trinidad (Ramtahal et al. 2019), the United States, and Australia (McLaughlin 2016), most likely due to a decrease in soil acidity. Ramtahal et al. (2019) found that while these materials were effective in reducing Cd levels in wheat (Tlustoš et al. 2006), rice (Chen et al. 2018), and lettuce (Woldetsadik et al. 2016), they were ineffective in cocoa at field level. The use of soil microorganisms capable of adsorbing, bioaccumulating, and biotransforming Cd has also been investigated as a

promising tool for the bioremediation of this element in cocoa-cultivated soils (Bravo and Braissant, 2022). Cadmium-tolerant bacteria such as *Pseudomonas aeruginosa* (Chakraborty and Das 2014), *Serratia spp.* (Sarma et al. 2016), *Halomonas spp.* (Siripongvutikorn et al. 2016), *Enterobacter spp.* (Chen et al. 2016), and filamentous fungi such as *Aspergillus spp.*, *Paecilomyces spp.*, *Microsporum spp.*, *Cladosporium spp.* (Mohammadian et al. 2015), and *Trichoderma spp.* (Cacciola et al. 2015) have shown the potential to bioaccumulate Cd.

Another option for preventing Cd bioaccumulation in cocoa is by breeding new varieties that are less prone to Cd uptake and using rootstocks with low Cd uptake in grafting (Maddela et al. 2020). Ullah et al. (2018) believe that a gene-based approach to the selection of low-Cd-uptake cocoa cultivars is possible because *T. cacao* has evolutionarily conserved Cd-uptake proteins. According to Maddela et al. (2020), selected genotypes reduce Cd uptake by rootstocks and have a low propensity to transfer Cd from rootstocks to leaves and fruits. In terms of using postharvest treatment to reduce Cd levels in cocoa beans, it is unclear how the fermentation process affects the Cd content of cocoa beans. This is because fermentation produces many acids, and Cd has high solubility and bioavailability in acidic environments (Abinandan et al. 2019). More research is needed to determine how post-harvest treatments like pod storage, fermentation, and drying affect Cd levels in cocoa beans.

Effect of climate change

Climate change has emerged as one of the most devastating global environmental threats, with far-reaching and unavoidable ramifications for cocoa farming and the livelihoods of millions of smallholder farmers. Climate change has a significant impact on cocoa production, and cocoa production has substantial climate change implications, making it an important policy formulation issue at all levels of development: local, national, and international. As the climate changes, cocoa-producing regions are expected to become more vulnerable to adverse weather conditions (Anning et al. 2022). According to Ceccarelli et al. (2021) and Igawa et al. (2022), existing cocoa-producing areas in West Africa, as well as the Brazilian and Peruvian Amazon, are expected to become less suitable for cocoa production by 2050 if no adaptation measures are taken. Läderach et al. (2013) developed a model that predicts that some current cocoa-producing areas in Côte d'Ivoire (Lagunes and Sud-Comoe) will become unsuitable for cocoa production, necessitating crop change, while other areas will require agronomic management adjustments. The model also predicted that climatic suitability for cocoa cultivation would improve in other areas (the Kwahu Plateau in Ghana and

southwestern Côte d'Ivoire). The study by Läderach et al. (2013) proposed site-specific strategies to reduce cocoa farmers' and the sector's vulnerability to future climate change.

Cocoa trees are extremely sensitive to climate change, and according to Hütz-Adams et al. (2022), the effects of climate change on cocoa productivity are already a reality in all cocoa-producing countries. The incidence of droughts and increased rainfall, temperature and solar radiation changes, and increased humidity as a consequence of climate change have an immediate impact on the overall health of the trees, their disease incidence, and their ability to set flowers and produce fruit (Hütz-Adams et al. 2022). Cocoa requires favourable climatic conditions at all production stages, including high temperatures, precipitation, and humidity (Kosoe and Ahmed 2022). Among the climate variables, Kosoe and Ahmed (2022) believe that rainfall has the greatest impact on cocoa yield, making the crop vulnerable to soil water scarcity. Seasonal pattern disruption, inconsistent rain, rising temperatures, and droughts are all potential climate risks to cocoa production. Atmospheric and soil droughts caused by Harmattan are seasonal hazards in West African cocoa agroecosystems that increase soil and air–water stress as evaporative demand rises due to higher wind speeds and lower air humidity, resulting in wide temperature differentials between day and night that are likely to harm cocoa production in West Africa (Della Sala et al. 2021). Cocoa production in wet areas, such as Latin America, is also vulnerable to climate change-related excess water, which causes soil-nutrient leaching and fungal diseases (Climate-Smart Cocoa 2022). In Ecuador, for instance, the rapid recurrence of unusually heavy rainfall in some coastal regions is expected to threaten up to 60% of current cocoa growing areas, leading to cocoa cultivation migrating to higher altitude areas away from the coast (Hütz-Adams et al. 2022).

Cocoa is produced by 5–6 million smallholder farmers worldwide, producing below cocoa yield potentials, and without a minimum living income (Fountain and Huetz-Adams, 2020; Bermudez et al. 2022). Climate change is projected to worsen these concerns by increasing climatic stress, which will harm cocoa-producing regions through rising temperatures, shifts in rainfall patterns, and more intense and frequent drought events (Malek et al. 2022). In the absence of adaptation measures, particularly, in the major producing countries, climate change will increase the vulnerability of cocoa production and disrupt global cocoa supplies, with knock-on effects on the economies of cocoa-producing countries, farmer livelihoods, and businesses across the cocoa value chain (Läderach

et al. 2013; Parra-Paitan et al. 2024). This would also have potential repercussions for forests and natural habitats as cocoa-growing regions expand, shrink or shift (Läderach et al. 2013). A recent decrease in cocoa production in West Africa for the 2023–2024 season, particularly in Côte d'Ivoire and Ghana, has resulted in a global shortfall and an increase in cocoa bean and cocoa-related product prices (Ritchie 2024; The Conversation, 2024). One climatic factor responsible for this shortfall is the impact of the El Niño weather phenomenon, which has caused drier weather in West Africa and climate change is exacerbating these conditions. Ritchie (2024) noted that the biggest impact of the El Niño weather phenomenon has not been the temperature itself but the impacts of rainfall extremes on disease outbreaks. West Africa experienced extreme wet conditions late last year, driving an outbreak of black pod disease. This extreme rain was followed by extremely dry conditions, which has helped the spread of the cocoa swollen shoot virus disease. Kosoe and Ahmed (2022) reported that climate change hastens the spread of swollen shoot virus disease and black pod disease, altering the crop's resistance to disease and pest infestation.

Cocoa production has been identified as a driver of climate change, in addition to being vulnerable to its effects (Lahive et al. 2019; Parra-Paitan et al. 2024). Cocoa farming is associated with deforestation in the majority of cocoa-growing regions (Bermudez et al. 2022). Deforestation is often associated with increased temperature and reduced rainfall, contributes significantly to GHG emissions (Vervuurt et al. 2022; Parra-Paitan and Verburg 2022), and destroys habitats, reduces biodiversity and degrades soil (Sassen et al. 2022). Cocoa cultivation increases total global warming potential (GWP) by three to four times due to land-use change (Aliouche 2023). Trees sequester carbon dioxide, and their loss can result in a net output of carbon dioxide into the atmosphere as well as a reduction in the concentration of carbon sinks. Important measures have been proposed to reduce the cocoa sector's vulnerability to climate change. These include breeding more temperature- and drought-resistant cocoa varieties and distributing them to farmers (Läderach et al. 2013; Schroth et al. 2016; Lahive et al. 2019), promoting shade trees in cocoa farms, and policies incentivizing the intensification of cocoa production on existing farms where future climate conditions allow, as well as establishing new farms in already deforested areas (Schroth et al. 2016), change in farm management practices such as planting the cocoa trees at a wider spacing to reduce their water needs per unit area, although this may also imply a greater effort for weed control as well as, initially, lower per-hectare yields (Läderach et al.

2013) and the establishment of early warning systems (Oduro et al. 2024).

Impact of COVID-19 and Russian-Ukraine War

The period from 2020 to 2023 witnessed major global events such as the COVID-19 pandemic and the ongoing conflict between Russia and Ukraine, with a concomitant impact on the cocoa industry. The COVID-19 pandemic had a significant impact on both the upstream and downstream cocoa sectors, as the majority of cocoa is exported to the chocolate confectionery industry. The pandemic hampered the availability of labour, inputs, and extension services, as well as the logistics of transporting dried fermented cocoa beans to markets (Global Agriculture and Food Security Program, 2021). Shipping disruptions to importing countries increased transportation costs and impacted product quality and availability. For example, between 2017 and 2021, European imports directly sourced from cocoa-producing countries decreased at a year-on-year rate of 0.6% (CBI, 2022). This drop can be attributed to the global pandemic slowing the cocoa and chocolate markets, resulting in a significant drop in direct imports in 2020 (CBI, 2022). The economic consequences of the pandemic, as well as government public health measures such as quarantines, are linked to a decline in demand for chocolate, as it is regarded as a non-essential item during the pandemic (Global Agriculture and Food Security Program, 2021; Tridge 2021). However, as governments' restrictions on people's movements have been lifted and consumers have resorted to buying chocolate in bulk and online, the situation has changed.

The ongoing Russia-Ukraine conflict is expected to have an impact on yields as trade sanctions have reduced fertilizer availability and the sharp price increases have made it difficult for some cocoa farmers to purchase inputs (Confectionery Production, 2022; ICCO, 2022). Cocoa is a fertilizer-intensive crop, so high fertilizer prices as a result of the conflict contributed to a lower-than-usual crop in 2022 (Semuels 2023). The Ghana Cocoa Board, Ghana's governmental regulator of the cocoa sector, has predicted that the conflict between Russia and Ukraine will have a negative impact on cocoa production in Ghana for at least two years, primarily due to fertilizer production and importation (Petetsi 2022). The conflict is also expected to have an impact on the cocoa bean market, as both countries are importers of cocoa beans (Bermudez et al. 2022).

Way forward towards sustainability

In the face of numerous cocoa production challenges such as pests and diseases, poor soil fertility, deforestation, irregular rainfall in the context of climate change,

global pandemic, and conflict, it is important to intensify the production of high-quality cocoa on existing farmlands in producing countries. Sustainable intensification of cocoa is required to ensure the preservation of other ecosystem services and the natural resource base on which cocoa cultivation is based, as well as to improve it for future generations. Sustainable intensification (SI) of agricultural products is gaining prominence in policy discussions and is at the forefront of food security policies to meet rising food demand while conserving land and other resources (Smith et al. 2017).

SI is defined as increasing the quantity and quality of cocoa beans produced by smallholder farmers per unit of land in a cost-effective manner to increase profitability and improve farmers' livelihoods, protect the environment (including the soil), conserve biodiversity and other agro-ecological services to ensure future generations can cultivate cocoa. SI is a practical approach to meeting the rising global demand for cocoa while minimizing environmental impact and improving the livelihood and social well-being of cocoa farmers. According to Boeckx et al. (2020), the SI of cocoa systems should include policies such as poverty alleviation and climate change mitigation. Indicators of SI can be divided into four categories: productivity, economic sustainability, environmental sustainability, and social sustainability (Fig. 7).

Cocoa productivity and quality

Cocoa productivity is defined as the amount of dried, fermented cocoa beans produced per unit of land and it includes the quality of the beans produced. Associated with productivity indicators in SI is the yield gap, or the difference between the actual yield of cocoa and

the potential yield (Tittone 2013). The potential yield is the amount of dried fermented cocoa beans that could be obtained under current soil conditions, water availability, solar radiation, and temperatures if all nutrient stresses and pest pressures were removed (Smith et al. 2017). Cocoa cultivation is predominantly carried out by smallholder farmers, with an average global yield of around 492 kg ha⁻¹ of fermented dry beans (FAOSTAT, 2023). While West African countries lead in total cocoa production, the average yield (productivity) is very low, around 300–600 kg ha⁻¹ (Asante et al. 2021), compared to potential yields of around 5000 kg ha⁻¹ under rainfed conditions (Zuidema et al. 2005) and over 3000 kg ha⁻¹ achieved in experimental trials (Appiah et al. 2000). This indicates a yield gap (the difference between potential and actual yields) of about 80–95% on existing cocoa farms. Reducing the current cocoa yield gap in the major cocoa-producing regions is crucial to helping improve the livelihoods of smallholder cocoa farmers.

Several factors have been identified to contribute to the high yield gaps in cocoa-producing regions. These include inadequate farm management practices, erratic rainfall patterns, temperature increases in the context of climate change, limited availability of improved planting material from national breeding programs, pests and diseases, low soil fertility, low plant density, inadequate shading levels, and ageing cocoa farmers and plantations (Asare et al. 2017; van Vliet and Giller 2017; Abdulai et al. 2020; Asante et al. 2022; Tosto et al. 2023). Establishing strategies to sustainably close current yield gaps requires a quantification of the relative contributions of yield-limiting factors as well as a thorough understanding of how cocoa responds to environmental conditions

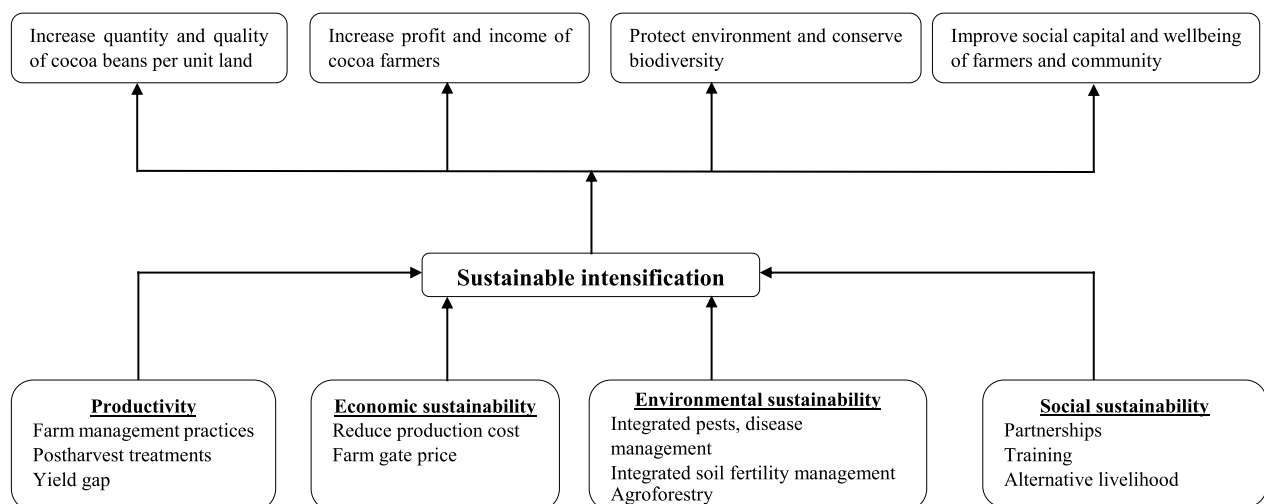


Fig. 7 Operational concepts of sustainable intensification (SI) of cocoa production; boxes beneath represent the concepts of SI with the measures involved in each concept; boxes above represent the outcomes of SI. Source: (Authors own construct)

and farm management practices. Although several studies have used yield gap analysis to quantify the contribution of yield-limiting factors (Abdulai et al. 2020; Asante et al. 2022), little is known about cocoa trees' responses to farm management practices and how these practices should be adapted to different climatic conditions and cropping systems (Tosto et al. 2023).

SI strategies to increase cocoa productivity in producing areas should include soil fertility programs that prioritize improving soil fertility parameters using locally available resources, such as organic resources. Diagnostic yield response trials on farmers' fields in cocoa-producing regions and under various shade management regimes should be conducted to test materials such as composted cocoa pod husks, household organic residues, compost, and green manure to develop an integrated soil fertility management (ISFM) plan. Most cocoa breeding programs focus on individual yield-limiting factors at a time (Tosto et al. 2023). Breeding programs should combine the yield-limiting factors to develop varieties that are resistant to all of the yield-limiting factors simultaneously. Again, cocoa yield is pollination-limited (Toledo-Hernández et al. 2023), where artificial or hand pollination has been found to increase yield compared to natural or insect pollination (Wongnaa et al. 2021). Studies by Toledo-Hernández et al. (2023) reported hand pollination by as little as 10% tripled fruit set and mature fruits in Brazil's agroforests. Earlier studies by Toledo-Hernández et al. (2020) also observed that 13% hand pollination, but neither fertilizer nor pesticide application, increased yield by 51%, while 100% hand pollination increased yield by 161%. More research on cocoa hand pollination is needed, and Toledo-Hernández et al. (2023) suggest that these studies should focus on a wide range of agroforestry, tree grafting, and innovation strategies in all major production regions to capture the long-term variability of hand pollination as a basis for scaling up hand pollination for sustainable cocoa production globally.

Despite several recommendations for best farm management practices and the use of improved cocoa varieties to control pests and diseases and increase cocoa productivity, farmers have been slow to adopt them. According to research, the adoption of agricultural technologies is influenced by a variety of factors, including socioeconomic, environmental, and local perceptions (Amon-Armah et al. 2021). Future cocoa sustainability research should focus on the factors that influence farmers' adoption of recommended farm management practices and improved cocoa varieties. Finally, cocoa is a woody perennial crop with a multi-decade production cycle, structural heterogeneity, and numerous ecological interactions within the system (Tosto et al. 2023). Thus, addressing the cocoa productivity gap through

agronomic or breeding experiments is time-consuming and costly. Tosto et al. (2023) proposed using cocoa simulation models that combine agroecological, physiological, and farming system knowledge with experimental and observational data to estimate cocoa yields, resource-use efficiency, and ecosystem services to supplement and inform experimental and on-farm research.

The age of cocoa farmers and cocoa trees are also important factors in productivity. The age of the cocoa tree is frequently regarded as a critical factor in reducing productivity. Given that yield rate increase peaks at 18 years after planting and begins to decline between 20 and 30 years (Binam et al. 2008; Wessel and Quist-Wessel, 2015), having too many overaged cocoa trees significantly reduces the potential cocoa yield on many farms. Farmers should be encouraged to replace old, unproductive cocoa trees with improved cocoa varieties that are resistant to yield-limiting factors on the same farmland. Again, cocoa cultivation is a low-technology venture that requires physical strength. Cocoa farmers in most major producing countries are relatively old, with an average age of more than 50 years (Rikolto 2023; Fairtrade, 2024), which may have an impact on physical strength and, thus, productivity. Access to land, finance, and insufficient agro-economic education or skills training for young farmers have been identified as the greatest challenges for young people to engage in cocoa farming (Löwe 2017). Linked to these challenges are the perceptions young people have about cocoa farming. According to Löwe (2017), young people see the older farmers working hard for little return and, thus, do not recognize the profitable livelihood potential in the cocoa sector. Policies and incentive packages that address access to land and finance should be formulated to encourage young people into cocoa farming.

SI, once again, entails the production of high-quality dried fermented cocoa beans for industrial use. The industry expects big beans with mass > 1.0 g and uniform size, moisture content of 6–8%, pH of 5.0–6.0 for improved flavour profile, well-fermented beans with > 60% brown beans, no slaty beans, high cocoa butter content (> 50%), with $\leq 1.75\%$ free fatty acid content of the cocoa butter (Afoakwa et al. 2008; CAOBISCO-ECA-FCC, 2015). Factors that could influence these quality attributes of cocoa beans include the genetic make-up and origin of cocoa, bean chemical composition, environmental conditions (such as rainfall and temperature), age of the cocoa tree, farm management practices, and postharvest treatments such as fermentation, and drying (Afoakwa 2016; CAOBISCO-ECA-FCC, 2015). Several studies on the impact of cocoa bean genotype, environmental conditions, and postharvest management on cocoa bean quality have been published (Quaraine

et al. 2012; Rodriguez-Campos et al. 2012). However, no research has been published on the impact of farm management on cocoa bean quality, such as weed control, pruning, fertilizer application, control of capsids and black pod disease, and soil quality, indicating a research gap that needs to be filled. Afoakwa (2014) noted that the factors responsible for the production of high-quality cocoa beans including cocoa genotype, environmental conditions, farm management and postharvest treatments are poorly understood thus, accounting for the inconsistent and unharmonized practices.

Cocoa cultivation is often characterized by the use of unsophisticated farm inputs and low technology resulting in far different production and farm management practices, soil fertility management and postharvest practices and strategies which are inconsistent and not harmonized (Afoakwa 2014; Asare et al. 2017; van Vliet and Giller 2017). Research is needed on the quality characteristics of cocoa beans produced from different farms under different farm management practices, soil quality and postharvest management strategies to ascertain practices which best influence cocoa bean quality. Findings from such research will aid policy formulation in harmonizing production and postharvest management to ensure sustainable production of high-quality cocoa beans.

Environmental sustainability

Cocoa cultivation harms the environment in two ways: deforestation and contamination from the use of chemicals to control weeds, pests, and diseases and improve soil fertility (Perez-Neira et al. 2020; Parra-Paitan et al. 2024). Cocoa is traditionally grown on fertile forestland with a thin forest canopy after the vegetation is cleared (Sassen et al. 2022). As yields begin to fall due to smallholder farmers' inability to revitalize their plantations, the vicious cycle of forest clearing resumes, resulting in deforestation. Again, current global cocoa management practices rely heavily on the use of inorganic fertilizers and traditional crop protection products to improve soil fertility and control weeds, pests, and diseases. Long-term and improper use of inorganic fertilizers and traditional crop protection products have been linked to human health risks, as well as environmental and biodiversity damage (Dhankhar and Kumar 2023). These chemicals have the potential to kill organisms that improve soil fertility while also polluting water bodies near cocoa farms.

Ending deforestation and promoting sustainable agroforestry systems and other environmentally friendly farming techniques are critical components of the cocoa sector's environmental sustainability. Environmental sustainability in cocoa SI thus, entails a multifaceted

approach to pest and disease control in cocoa fields, as well as conserving soil water, increasing soil nutrients, and preserving biodiversity. This holistic approach employs cost-effective and easily accessible materials to address targeted pests and diseases as well as soil fertility issues in producing areas in an effective and environmentally friendly manner. This includes integrated pest and disease management (chemical, biological, and physical control), integrated soil fertility management (using locally generated materials such as cocoa pod husks, household organic residues, compost, and green manure to improve soil fertility), and agroforestry (the planting of multi-purpose trees). The integrated management approach will contribute to reducing the use of chemicals and the cost of cocoa production, increasing the productivity, profit, and income of cocoa farmers, as well as protecting the environment for future generations. Studies have demonstrated the environmental and ecological benefits of agroforestry systems, which promote biodiversity (Marconi and Armengot 2020; Kouassi et al. 2021). Despite all of the benefits of cocoa agroforestry systems, lower cocoa yields remain one of the most significant barriers to the widespread adoption of diversified production systems (Niether et al. 2020). As a result, more research is needed into increasing cocoa yields in agroforestry systems through shade-tolerant variety breeding or adapted management practices to improve pollination rates.

Again, to achieve environmental sustainability in SI, cocoa traders and importers must take the lead in implementing zero-deforestation action in areas where cocoa is responsible for the majority of deforestation caused by agri-commodities. However, cocoa traders who source from areas where other commodities are major drivers of deforestation must link voluntary sustainability initiatives to public initiatives, land-based sector initiatives, and territorial initiatives (Parra-Paitan et al. 2024). Commodity traders, including multinational corporations such as Cargill, Olam, and Bunge, as well as many smaller businesses, are key actors in efforts to eliminate deforestation because they are active in the regions where commodities are produced and play an important role in achieving deforestation-free sourcing (Zu Ermgassen et al. 2022). In response to most conventional chocolate consumers' desire to know how much deforestation is embedded in their purchases (Carodenuto 2019), the global chocolate industry announced in 2017 their collective commitment to 'zero deforestation cocoa,' in which companies aim for full supply chain traceability to eventually end deforestation in cocoa-growing regions (World Cocoa Foundation [WCF], 2020). This zero-deforestation commitment (ZDC), known as the Cocoa and Forests Initiative (CFI), represents a new era of sustainable sourcing practices in

global agriculture commodity production and trade, with corporations pledging voluntary sustainability measures for their supply chains (Furumo and Lambin 2020; Carodenuto and Buluran 2021).

Economic sustainability

Central to cocoa SI are the smallholder farmers responsible for over 90% of global cocoa production. Among cocoa industry sectors, the downstream sector, particularly final manufacturers and retailers, receives a share of 79.4%, while cocoa farmers receive 6.6% (Fountain and Hütz-Adams, 2015). Despite the global chocolate industry's high value of US \$130 billion (World Economic Forum, 2020), farmers who produce cocoa, the one key ingredient on which the chocolate industry relies, remain poor and have low living standards (Boysen et al. 2023). Providing a living income for cocoa farmers and a fair distribution of the value generated along the supply chain are critical components of the cocoa sector's economic sustainability. Economic sustainability thus entails increasing profit margins in cocoa farming, particularly for cocoa farmers, to improve their livelihoods. A key component of economic sustainability in cocoa is increased cocoa productivity. Low productivity and poverty undermine SI in cocoa because they are the primary causes of environmental sustainability and child labour issues associated with cocoa farming (Fountain and Hütz Adams, 2020). Reducing cocoa production costs through improved cocoa varieties that are resistant to yield-limiting factors, integrated pest and disease management, integrated soil fertility management, and agroforestry would all contribute to increased productivity and profitability in cocoa production.

Another key component of economic sustainability in cocoa SI is the farm gate price paid to cocoa farmers for the sale of cocoa beans (Anim-Kwapong and Frimpong 2008). Farm gate price influences farmers' decisions to invest in cocoa farms by implementing recommended farm management practices that require the purchase of external inputs. High farm gate prices that cover or exceed production costs increase cocoa farmers' profit margins and income. This will encourage farmers to improve farm management practices such as controlling pests and diseases, applying fertilizer, weeding and mistletoe removal, pruning, thorough harvesting, fermentation, and drying. Policies such as the cocoa Living Income Differential (LID), which was jointly introduced by the two largest cocoa producers, Côte d'Ivoire and Ghana, to address cocoa farmers' incomes (Boysen et al. 2023), should be implemented in all cocoa-growing areas, but tailored to each cocoa-growing area. Such policies should be continuously monitored and evaluated through research to ensure the objectives are met while

avoiding unintended negative social and environmental consequences.

Again, the current pricing of cocoa beans (farm gate price) in most producing countries focuses only on the quantity (kg) of beans produced and does not reward quality. In this regard, quality should be rewarded by paying a little above the approved farm gate price (premium) to farmers who produce well-fermented dried cocoa beans of high quality. This will encourage cocoa farmers to invest resources in implementing recommended best practices to produce more high-quality beans. This would also necessitate a rapid (possibly non-invasive) assessment of cocoa bean quality before purchase. Currently, the cocoa bean cut test is the simplest and most widely used method for assessing the quality of a random sample of beans from a batch by visual inspection (Kongor et al. 2013), and it is typically performed at the port level before export rather than the farm gate. Despite its simplicity and use by trained personnel, this method is laborious, time-consuming, subjective, and difficult to standardize (Kongor et al. 2013). There is a need for a quick and accurate technique for grading dried fermented cocoa beans at the farm gate that is low-cost and widely standardized. Araujo et al. (2014) proposed a cocoa bean quality index for determining cocoa bean quality. Teye (2022) also used mini-shortwave spectroscopic techniques and multivariate statistical analysis to test intact cocoa beans at farm gates for quality control in Ghana. Teye's (2022) study, however, focused solely on moisture content and pH in determining cocoa bean quality. More research is required to define cocoa bean quality, standardize these quality components, and incorporate them into a non-destructive tool for estimating cocoa bean quality at the farm gate.

Social sustainability

Recognized as an important component of sustainability, social sustainability is linked to 11 of the 17 Sustainable Development Goals (SDGs) of the United Nations (Wang and Ke 2024). Bostrom (2012) defines social sustainability as human welfare, which includes quality of life, social justice, social cohesion, cultural diversity, democratic rights, gender issues, human rights, participation, social capital development, and human capability. Social sustainability, a critical component of cocoa SI, entails increasing the social capital and well-being of cocoa farmers and their communities, with a focus on achieving improved human well-being and quality of life. It includes collaborations between cocoa sector managers in producing countries, cocoa/chocolate industries, and non-governmental organizations (NGOs) to train cocoa farmers and extension officers on farm management and postharvest management practices, the provision of basic

amenities (schools, healthcare, and water) in farming communities, and the creation and support of alternative livelihoods (subsistence food production).

Farmer poverty (CBI, 2023) and labour-related issues, particularly the use of child labour (Fountain and Hütz-Adams, 2020) have been identified as the most significant social sustainability challenges in the cocoa industry. There is a high prevalence of the use of child labour on cocoa plantations in cocoa-producing countries in West Africa (Fountain and Hütz-Adams, 2020). In Latin America, however, labour rights and freedom of association are the pressing social challenges (Hütz-Adams et al. 2022). Thus, responsible business conduct by key stakeholders in the cocoa value chain is required to eliminate farmer poverty, child labour, and other labour-related issues and ensure the cocoa sector's social sustainability. Increased cocoa productivity and farm gate prices would help to alleviate poverty and improve farmers' livelihoods. To increase cocoa productivity and profitability, governments, NGOs, and cocoa/chocolate industries must increase capacity-building for cocoa farmers in areas such as good farm management and postharvest practices, integrated pest and disease management, integrated soil fertility management, and agroforestry.

Also, to alleviate the effects of low profitability in cocoa production and enhance the livelihood of cocoa farmers, policies that provide technical support and training for the integration of alternative livelihoods and other income opportunities in parallel with cocoa production (subsistence food production such as intercropping food crops on cocoa farms, livestock production, valorization of cocoa waste, and cash-earning activities such as handicrafts, small businesses) should be formulated. According to Franzen and Borgerhoff (2007), farm diversification may be an effective way to improve farmer security while discouraging farmers from abandoning or planting cocoa due to price fluctuations, thereby reducing the use of new forest areas for cocoa production. To eradicate child labour in cocoa-producing countries, there must also be improvements in the quality and accessibility of general education, including well-planned school feeding programs, education on children's rights, and decent work in agriculture, as well as alternative livelihood and development opportunities for children.

Conclusion

Cocoa is a valuable agricultural commodity with global economic significance. Millions of people worldwide consume cocoa products such as chocolate and beverages. Despite cocoa's economic and health benefits, the industry, particularly the downstream sector, faces numerous challenges, including pests and diseases, poor soil fertility management, cadmium toxicity, prolonged

dry seasons, increased rainfall, extreme temperatures, and the ongoing Russian-Ukrainian conflict. These factors have an immediate impact on cocoa productivity, the environment, and the livelihoods of smallholder farmers and producing countries. Cocoa production is not only affected by climate change, but it also contributes to it. Cocoa cultivation is associated with deforestation and biodiversity loss in most producing areas, raising the total global warming risk.

In most major producing countries, smallholder cocoa farmers have low productivity, which leads to low income and widespread poverty. Again, due to the low technology and input base, cocoa production in the major producing countries is characterized by vastly different production and postharvest management strategies that are inconsistent and not harmonized, resulting in variations in cocoa bean quality. The economic and health benefits of cocoa are dependent on the sustained intensification of high-quality cocoa bean production in all producing areas. More concerted efforts are needed to increase cocoa productivity on existing farmlands while also preserving other ecosystem services and the natural resource base that cocoa cultivation relies on, as well as improving it for future generations.

To achieve sustainable intensification in cocoa production, cocoa breeding programs should combine yield-limiting factors to create varieties that are resistant to all of them at the same time. Cocoa SI programs should prioritize the integrated management of pests and diseases, heavy metals like Cd, and soil fertility (using locally generated materials such as cocoa pod husks, household organic residues, compost, and green manure to improve soil fertility). These approaches would help to reduce chemical use and cocoa production costs while increasing productivity and environmental protection. Despite the ongoing debate about cocoa monoculture versus agroforestry systems as the best cropping system, several studies have shown that agroforestry systems have environmental and ecological benefits that promote biodiversity. Lower yields are the most significant disadvantage of cocoa agroforestry. In this regard, more research should focus on increasing cocoa yields in agroforestry systems under different shade management strategies as well as breeding shade-tolerant varieties. Again, a thorough understanding of the impact of various farm management and postharvest practices on cocoa bean quality is required to ensure standardized and harmonized management strategies, which are necessary for the consistent production of high-quality beans each season.

Smallholder farmers play a central role in sustainable intensification. Increased efforts are required to improve cocoa farmers' incomes through policies such as the Living Income Differential and quality-based incentives.

Cocoa industry stakeholders, including governments, cocoa traders, importers, and processors, must take the lead in implementing zero-deforestation measures in areas where cocoa is responsible for most agri-commodity-related deforestation. However, cocoa production alone will not help farmers get out of poverty. Policy directives encouraging collaboration among sector managers, non-governmental organizations, and the cocoa and chocolate industries should be developed to provide training and social interventions such as alternative livelihoods for farmers and farming communities. These would help to diversify farms, improve farmer security, and discourage farmers from abandoning or planting cocoa due to price fluctuations, thereby reducing the need for new forest areas for cocoa production.

Abbreviations

ICCO	International cocoa organization
FAO	Food and agriculture organization
FFC	Fine flavour cocoa
WCF	World cocoa foundation
SI	Sustainable intensification
GDP	Gross domestic product
GHG	Greenhouse gas
GWP	Global warming potential
GAP	Good agricultural practices
CSSVD	Cocoa swollen shoot virus disease
VSD	Vascular streak dieback
ZDC	Zero-deforestation commitment
CFI	Cocoa and forests initiative

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